The Rise of Verbal Reinforcement Learning: A Survey

Anonymous ACL Submission

Abstract

Natural language is emerging as a primary feedback channel for improving language agents, capable of conveying intent, preferences, and causal structure in forms interpretable by both humans and modern language models. We call this paradigm Verbal Reinforcement Learning (VRL) and offer the first unified account of it, with explicit inclusion criteria that separate it from RLHF, instruction following, and language-conditioned RL. We organize the field around a single axis, *when* verbal feedback takes effect in an agent's lifecycle and *what* it modifies, yielding three pillars: (1) **Language as Grounding Signal**, where language defines the task itself by specifying goals, states, and reward structures; (2) **Language as Deliberative Feedback**, where natural language guides reasoning at test time without the need to update model parameters; (3) **Language as Learning Signal**, where language-based feedback shapes model parameters through training. Within each pillar, we synthesize representative work, distinguish key subcategories of approaches, and outline the distinct role language plays in shaping agent behavior. Together, this taxonomy shows how verbal reinforcement is reshaping agent development, while also defining the challenges and opportunities for building more capable and aligned agents.

1 Introduction

Classical reinforcement learning has achieved remarkable successes by coupling agents with well-specified environments, action spaces, reasoning mechanisms, and task-specific reward functions. This framework has enabled agents to master Atari games (Mnih et al., 2015), defeat world champions at Go (Silver et al., 2016), optimize chip placement (Mirhoseini et al., 2021), and control complex physical systems (Degrave et al., 2022). However, each success rests on heavy domain-specific engineering: the environment must be formalized, the

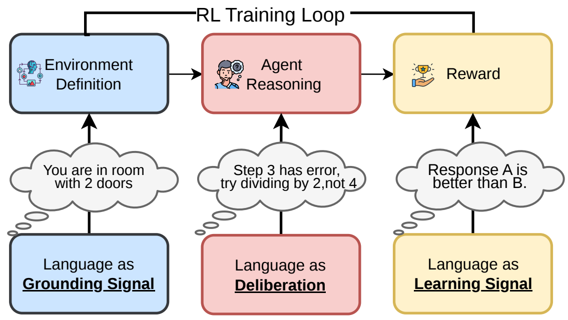

Figure 1: Language plays three complementary roles in agent's development lifecycle: grounding the environment, scaffolding intermediate reasoning, and providing learning signal for further training.

interaction space defined, and the reward function must closely capture the intended behavior (Vamplew et al., 2021). Designing such specifications is a long-standing bottleneck (Amodei et al., 2016), especially in settings where tasks and goals are ambiguous, context-dependent, and difficult to encode directly.

The emergence of large language models (Brown et al., 2020) has introduced an alternative supervisory channel: natural language. Instead of requiring all task intent to be predefined, verbal feedback enables on-the-fly communication of corrections, preferences, and constraints in a manner that agents are increasingly able to interpret and implement. Modern large language models, particularly frontier models (Achiam et al., 2023), can accurately describe a wide range of phenomena. These capabilities include describing environments (Wang et al., 2023a), identifying bugs in code (McAleese et al., 2024), critiquing generated outputs (Madaan et al., 2023), and translating human preferences into actionable training signals (Hong et al., 2024a; Meng et al., 2024). Similar to how self-supervised learning (Devlin et al., 2019; Chen et al., 2020) expanded the use of unlabeled data in model development, verbal feedback is emerging as an important

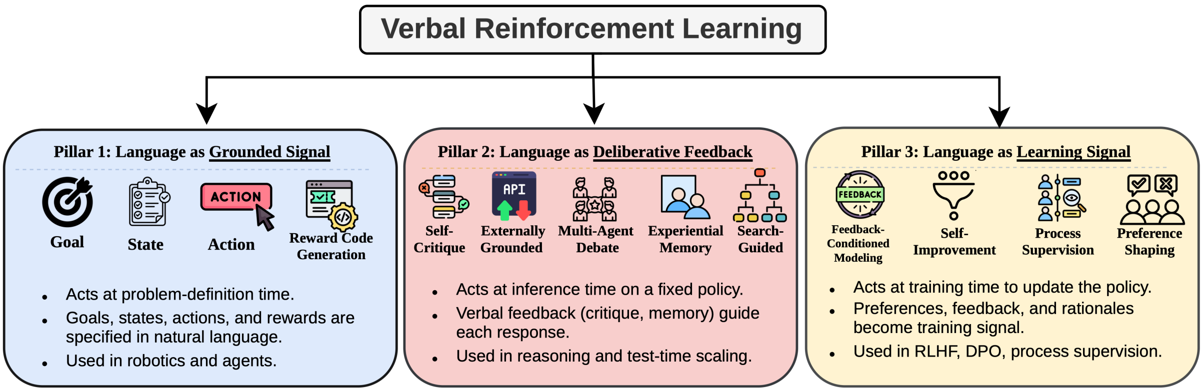

Figure 2: Our proposed taxonomy of Verbal Reinforcement Learning. Language serves three distinct functional roles: *grounding signal*, which structures environment interaction; *deliberative feedback*, which operates at inference time to refine reasoning; and *learning signal*, which drives parameter updates at training time.

source of supervision for language-model-based agents.

We refer to this family of methods as **Verbal Reinforcement Learning (VRL)**, and we define it by two inclusion criteria: *(i) the feedback is expressed in natural language, whether self-generated, provided by a human, or produced by an external tool or model, and its linguistic structure is functionally consumed by the agent, rather than being collapsed to a scalar or preference bit before use; and (ii) the feedback participates in an improvement loop, that is, an output revision, a memory write, or a parameter update.* Both criteria are required, and they fence VRL off from adjacent paradigms. Plain instruction following satisfies neither, since no improvement loop exists. Vanilla RLHF and DPO satisfy only the second, because the verbal judgment is reduced to a preference bit before learning; we discuss them only as the scalar endpoint of the compression spectrum in Section 5. Language-conditioned RL treats language as an input to the policy rather than as feedback on its behavior. One boundary case the criteria settle: language consumed at specification time to construct a reward function, as in reward-code generation, satisfies criterion (i) even though the constructed function later emits scalars. The term was introduced by Shinn et al. (2023) to describe self-reflective agents (specific instance of what we call deliberative feedback); we deliberately generalize it to any method meeting the two criteria above, and we state this generalization explicitly to avoid conflation with the original, narrower sense.

Early results demonstrate that verbal feedback can drive substantial improvements across diverse settings. Ahn et al. (2022) used language to ground high-level instructions in physical affordances, enabling real-world robotic planning. Shinn et al. (2023) achieved 91% pass@1 on HumanEval through verbal self-reflection alone, and Madaan et al. (2023) showed that iterative verbal critique yields substantial gains across diverse generation tasks. Ouyang et al. (2022) demonstrated that a 1.3B-parameter model trained on verbal preference judgments outperformed the 175B GPT-3 baseline, a 130× size disadvantage overcome by richer supervision. These are not isolated demonstrations and publications on arXiv referencing "self-refine," "self-correction," "verbal feedback," and "language-feedback alignment" have grown from a handful in early 2020 to several hundred today, spanning multiple domains including coding assistants (Yang et al., 2024; Jimenez et al., 2024), scientific discovery (Bran et al., 2024; Lu et al., 2024), robotics (Wang et al., 2023a), mathematical reasoning (Hatamizadeh et al., 2026; Liu et al., 2026b; Shi et al., 2025a), clinical decision-making (Huang et al., 2025b; Zhou et al., 2025b) and education (Qian et al., 2025; Ahtisham et al., 2026).

These methods differ in their mechanisms (Figure 1): some operate during inference, others modify training data, and some redefine the task itself . However, they share a common characteristic: the use of natural language to guide, refine, and ground agent behavior. Yet no existing framework unifies them. Prior surveys have examined language-conditioned policies in text-based environments (Luketina et al., 2019), the bidirectional synergies between LLMs and RL (Pternea et al., 2024), LLM self-correction in isolation (Kamoi et al., 2024), automatic correction strategies (Pan et al., 2024), feedback integration for text generation (Fernandes et al., 2023), preference optimization (Kaufmann et al., 2023; Xiao et al., 2024), agent memory (Zhang et al., 2025e), test-time scaling (Zhang et al., 2025b), self-evolving agents (Gao et al., 2025; Fang et al., 2025), and reinforcement learning for LLM agents (Zhang et al., 2025a); Natural Language Reinforcement Learning (Feng et al., 2024) recasts RL components in language space but is a framework proposal rather than a survey. Table 3 in Appendix A positions our taxonomy against these works: they organize by mechanism or by agent capability, whereas we organize by the signal itself. In contrast, our survey focuses on the functional role of *verbal feedback* as a first-class signal for improving agent, organized along a single axis: *when* natural language enters the agent's lifecycle and *what* it modifies. We formalize this axis into a three-pillar taxonomy in Section 2. This paper makes following contributions:

- We present the first comprehensive survey of Verbal Reinforcement Learning, defining the paradigm through explicit inclusion criteria that separate it from RLHF, instruction following, and language-conditioned RL, and proposing a three-pillar taxonomy (Section 2) that organizes the field by *when* language acts in the agent lifecycle.
- Within each pillar, we synthesize representative methods, identify cross-cutting challenges and outline future directions for developing robust, capable and aligned Verbal RL agents.
- We give explicit decision rules for methods whose signals span pillars, including a hybrid-cases table (Table 1), so the taxonomy is applicable rather than merely descriptive.
- We distill practitioner guidance for choosing among verbal-feedback mechanisms (Appendix B) and position the survey against prior overviews (Appendix A).

We organize the paper as follows. Section 2 introduces the three-pillar taxonomy and a working example showing how the pillars operate together in a single coding agent. Sections 3–5 develop each pillar of the taxonomy in detail: language as grounding signal, as deliberative feedback, and as learning signal, respectively. Finally, section 6 presents cross-cutting insights and outlines future research directions, while section 7 provides the conclusion.

2 Taxonomy

We organize VRL methods along a single axis: *when* natural language enters the agent's lifecycle and, consequently, *what* it modifies. This yields three pillars, summarized in Figure 2 and synthesized in Table 2.

- **Pillar 1: Language as Grounding Signal** (§3) acts at *problem-definition time*, specifying the MDP (Sutton and Barto, 1998) itself: goals, states, actions, and rewards. A natural-language task description such as *"You are in a room with two doors"* defines what the agent perceives, how it can act, and what counts as success, all before any policy runs.
- **Pillar 2: Language as Deliberative Feedback** (§4) acts at *inference time*, refining outputs or reasoning traces through critique, memory, debate, or search, without updating parameters. A message such as *"Step 3 has an error; try dividing by 2, not 4"* redirects a single generation episode but leaves the model unchanged for future tasks.
- **Pillar 3: Language as Learning Signal** (§5) acts at *training time*, where verbal feedback is distilled into gradient updates that persistently reshape the policy. A judgment such as *"Response A is better than B because it addresses the edge case"* becomes a preference pair that alters model weights, affecting all subsequent interactions.

The key distinction is *persistence*: grounding defines the task space, deliberative feedback improves a single episode, and learning signal permanently reshapes the policy. We discuss each pillar in consecutive sections. For methods whose signals span pillars, we use an explicit assignment rule: *a method is placed by what its linguistic signal modifies at the moment the signal takes effect*. Table 1 applies this rule to the recurring hybrid cases and records their secondary roles instead of forcing each method into a single cell. Throughout, MDP vocabulary serves as an organizing lens rather than a claim that every method admits an MDP formulation; many language-agent settings are partially observable, non-Markovian, or lack explicit transition and reward functions, and we note where the formalism stops applying.

Table 1: Assignment rule applied to hybrid cases: placement follows what the signal modifies at the moment it takes effect; secondary roles are tagged, not suppressed.

Hybrid case	Primary	Secondary role
Reward-code generation (Ma et al., 2024)	P1	Constructed reward later trains a policy (P3)
Experiential memory (Shinn et al., 2023)	P2	Lessons persist across episodes, bordering long-term adaptation
Constitutional AI (Bai et al., 2022)	P3	Self-critique generates the revisions at inference (P2)
Feedback-conditioned fine-tuning (Luo et al., 2025)	P3	Critique text retained verbatim in the training context

Table 2: Overview of the three VRL pillars.

Axis	Grounding Signal	Deliberative Feedback	Learning Signal
When	Problem-definition time	Inference time	Training time
Modifies	Task, State, Action, Reward	Outputs or reasoning traces	Model weights, Training data
Persistence	Defines the task	Single episode (extendable via memory)	All future interactions
Methods	Environment specification	Prompting and critique	SFT, PPO, DPO
Challenges	Grounding gap; Compositionality	Feedback Quality; Compute Cost	Signal Compression; Filter Quality

Working Example: the coding agent - The three pillars are complementary, not competing. We emphasize that the pillars classify individual signals, not stages of a pipeline: no method is claimed to traverse a grounding-deliberation-learning cycle, and the example below shows a system in which signals of all three kinds coexist. Consider an agent (Wang et al., 2025d) tasked with resolving a GitHub issue. *Grounding* (Pillar 1) enters first: the natural-language issue description defines the goal, the repository structure specifies the state space, and the test suite implicitly encodes a reward function. At inference time, *deliberative feedback* (Pillar 2) drives the iterative loop: the agent generates a patch, executes the tests, treats traces and error messages as verbal feedback, critiques its own output, and revises. When successful

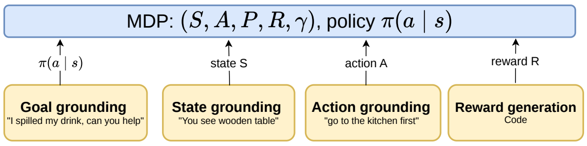

Figure 3: Grounding of MDP components through natural language. Each subcategory maps verbal feedback to a specific element of the MDP: goal grounding specifies the objective, state grounding represents observations as text, action grounding resolves instructions to executable behavior, and reward code generation compiles language into executable reward functions.

and failed trajectories are later collected and used to fine-tune the model via preference optimization, they become a *learning signal* (Pillar 3) that persistently improves the policy for future tasks. The same agent thus consumes verbal feedback at three distinct timescales, each producing a different effect. This example demonstrates the rationale for organizing by temporal scope rather than by feedback source (human, model, tool) or modality (text, scalar, code). Identical verbal feedback can yield fundamentally different effects depending on when it is consumed, and each temporal regime introduces distinct technical challenges, which we will discuss in later sections.

3 Pillar 1: Language as Grounding Signal

In this pillar, we discuss the role of language in providing grounding for each component of the task in which an agent operates.As illustrated in Figure 3, verbal feedback maps to specific MDP elements (Sutton and Barto, 1998): specifying what the agent should optimize (goals - §3.1), what it perceives (states- §3.2), how it can act (actions- §3.3), and what constitutes success (rewards- §3.4) (MacGlashan et al., 2017). This matters because many failures in language-based agents stem not from poor optimization but from poor grounding: feedback that cannot map to executable actions or valid reward conditions will not improve the agent. A common bottleneck across all four subcategories is the grounding gap. More detailed verbal specifications do not necessarily result in improved grounding. The critical factor is whether the mapping from language to MDP components is sufficiently precise for the agent to execute actions.

3.1 Goal Grounding

Goal grounding uses verbal feedback to specify what the agent should achieve, requiring parsing instructions into objects, relations, and target conditions (Chaplot et al., 2018), filtering plans by physical executability (Ahn et al., 2022; Driess et al., 2023), and decomposing goals into verifiable subgoals. Recent work extends this to multi-agent coordination (Zhang et al., 2025d), hierarchical decomposition via offline RL (Hu et al., 2025), and training from language specifications without manual reward functions (Li et al., 2026). The open challenge is compositional generalization: agents must follow goals assembled from familiar primitives in unfamiliar combinations. BabyAI (Chevalier-Boisvert et al., 2018) isolates this in gridworlds, and even state-of-the-art LLMs achieve only about 75% coverage under compositional constraints (Sakai et al., 2025), though compositional policy representations can improve sample efficiency on novel compositions (Cohen et al., 2025).

3.2 State Grounding

Where goal grounding defines what to achieve, state grounding defines what the agent perceives. Verbal state descriptions are most explicit in text-based environments (Yao et al., 2020; Côté et al., 2018), but appear whenever visual or physical states are summarized into language. Shridhar et al. (2020) show that agents trained on verbal state descriptions transfer to embodied 3D settings, demonstrating language as a modality-invariant state representation. Recent work includes closed-loop state feedback for robotic planning (Bhat et al., 2024) and scene-graph-based grounding (Huang et al., 2025a). A key challenge is information preservation, specifically whether the verbal abstraction retains sufficient signal for the policy to operate effectively.

3.3 Action Grounding

Given a goal and a perceived state, the agent must determine how to act. Action grounding resolves verbal instructions to specific skills, tool calls, or motor commands, requiring that instructions map to executable actions and that execution outcomes feedback as language to revise subsequent steps. Huang et al. (2022b) demonstrate this closed loop with Inner Monologue, while Sharma et al. (2022) show that short corrective utterances can amend mistaken steps without disrupting the rest of the plan. Recent systems have pushed toward tighter integration: multi-modal grounded replanning that corrects ambiguous instructions using visual cues (Kim et al., 2025), constraint-aware visual programming for proactive failure detection (Zhou et al., 2025a), corrective planning that handles physical, logical, and semantic errors across hierarchical control levels (Joublin et al., 2024), and visually grounded task-and-motion planning that combines LLM reasoning with physical feasibility (Zhang et al., 2025c). The central challenge is granularity alignment: the same instruction can refer to a temporally extended skill (“make coffee”), a discrete subgoal (“go to the kitchen”), or a motor command (“close the gripper”), each demanding a different language-to-control interface.

3.4 Reward Code Generation

The preceding subcategories ground objectives, observations, and actions in language. Reward code generation closes the loop by grounding success itself. This subcategory overlaps with Pillar 3 when generated rewards are used for training; we categorize it here because verbal feedback defines the reward function itself, becoming part of the MDP, and Table 1 records the downstream training role explicitly. Under our inclusion criteria, the linguistic signal is consumed at specification time, so reward-code generation satisfies criterion (i) even though the compiled function later emits scalar rewards. The distinguishing feature is compiling verbal descriptions into executable reward functions rather than using language as a direct learning signal. Ma et al. (2024) introduced Eureka, where an LLM generates and iteratively refines reward code based on training summaries, and Xie et al. (2024) extended this to dense reward shaping. This area has seen rapid recent progress: CARD (Sun et al., 2025a) introduces dynamic trajectory-based feedback to refine reward code without per-iteration RL training, PROF (Sun et al., 2025b) extends the paradigm to offline settings through preference optimization, and Zeng et al. (2024) show that reward functions can be generated from video demonstrations. To ensure effectiveness, verbal feedback must be compilable, executable, and accurately reflect the original natural-language intent.

4 Pillar 2: Language as Deliberative Feedback

With the task grounded, we turn to methods that refine reasoning within a single episode without updating model parameters, closely aligning with test-time compute scaling (Snell et al., 2024). As illustrated in Figure 4, five subcategories differ in where critique originates: the model itself (§4.1), external tools (§4.2), peer models (§4.3), past episodes

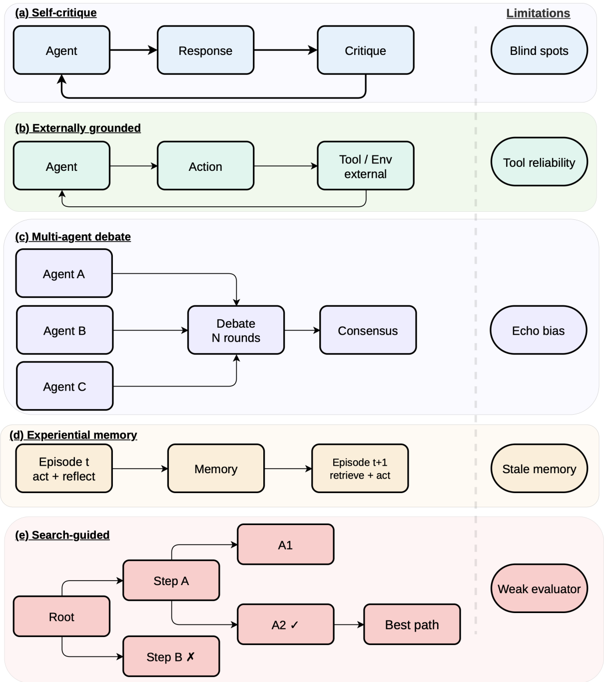

Figure 4: Inference-time verbal feedback mechanisms. Each subcategory exhibits a distinct feedback loop structure: (a) self-critique loops within a single model, (b) externally grounded critique incorporates verified tool or environment output, (c) multi-agent debate distributes critique across parallel model instances and converges to a refined output, (d) experiential memory persists lessons across episodes, and (e) search-guided deliberation explores and prunes multiple candidate paths.

(§4.4), or parallel search paths (§4.5).

4.1 Self-Critique

The simplest deliberative loop asks the model to critique its own output and revise accordingly. Self-Refine (Madaan et al., 2023) formalized this pattern, building on Constitutional AI's (Bai et al., 2022) demonstration that models can evaluate outputs against written principles. The primary challenge is circularity. Replacing self-generated critique with evaluation from a separately trained critic enhances reasoning (Paul et al., 2024), which indicates that feedback quality is the main bottleneck. Self-correction degrades when the critique source shares the generator's blind spots (Huang et al., 2024; Kamoi et al., 2024), especially in smaller models (Olausson et al., 2024). Mitigation strategies include Socratic decomposition into verifiable steps (Shi et al., 2025a) and parallel self-refinement across candidates (Wang et al., 2025c). However, when errors arise from gaps in the model's knowledge, internal feedback mechanisms cannot identify the missing information.

4.2 Externally Grounded Critique

External grounding addresses this circularity by anchoring verbal feedback in deterministic tool output (Schick et al., 2023). Verified signals, such as execution traces, unit test verdicts, search results, and API responses (Chen et al., 2024; Gou et al., 2024b), facilitate the generation of corrected outputs (Gou et al., 2024a; Yang et al., 2024; Wang et al., 2024b). Additionally, end-to-end reinforcement learning enables models to utilize such feedback more effectively (Gehring et al., 2024). The primary failure mode shifts from blind spots to trust asymmetry: the model receives accurate tool output but may misattribute the root cause of a reported trace. Practical limitations persist, including infrastructure overhead, poor scalability of latency, and the restriction of grounded critique to domains with a verifiable oracle.

4.3 Multi-Agent Critique / Debate

Multi-agent debate distributes verbal critique across parallel model instances, frequently assigning distinct personas to enhance perspective (Hong et al., 2024b). The exchange of critiques can improve reasoning even among identical models (Du et al., 2024), while the introduction of diverse roles further strengthens the quality of discussion (Chan et al., 2023). Communication topology is a critical factor; full-broadcast designs incur substantial token overhead (Choi et al., 2026), motivating the exploration of sparse alternatives (Li et al., 2024). Models may also be trained specifically for debate, enabling them to revise reasoning in response to conflicting opinions (Liu et al., 2026a). Adaptive heterogeneous frameworks have demonstrated improvements in mathematical reasoning (Zhou and Chen, 2025). However, the primary limitation is that distributing critique is only effective when participating models are genuinely diverse. When models share biases, the process yields redundant consensus (Estornell and Liu, 2024; Liang et al., 2024), and current debate methods do not consistently outperform simpler single-agent strategies (Choi et al., 2026).

4.4 Experiential Memory

Previous approaches regenerate feedback from scratch in each episode. In contrast, experiential memory distills reusable lessons into persistent storage, categorized as episodic (raw logs), semantic (distilled rules), and procedural (reusable skills)

(Zhang et al., 2025e). Although stored lessons persist across episodes, we classify memory by the moment of consumption: a retrieved lesson acts as deliberative feedback within the current episode, and Table 1 records the boundary with long-term adaptation. Reflexion (Shinn et al., 2023) stores raw verbal feedback, ExpeL (Zhao et al., 2024) extracts insights from paired successes and failures, and Voyager (Wang et al., 2023a) accumulates executable skills. Recent surveys have formalized these mechanisms (Du, 2026), while MemBench evaluates memory quality through downstream performance (Tan et al., 2025). However, persisting feedback can also perpetuate errors: poor-quality memories propagate mistakes, expanding memory stores require complex retrieval (Park et al., 2023), outdated knowledge can mislead in non-stationary environments (Zhang et al., 2025e), and adversarial injection may induce cross-session drift (Dong et al., 2026).

4.5 Evaluator Feedback in Search

The inclusion criterion for this subcategory is the verbal evaluator signal consumed during search, not the search procedure itself; decoding strategies that involve no linguistic feedback fall outside VRL. While previous approaches refine a single trajectory, these methods explore multiple candidate paths and employ verbal feedback to determine which paths to expand, prune, or abandon (Wang et al., 2025a). The Tree of Thoughts framework (Yao et al., 2023) samples several intermediate thoughts and evaluates them through self-assessment, thereby extending chain-of-thought prompting (Wei et al., 2022) with a tree-based search mechanism. Reasoning via Planning (Hao et al., 2023) and Graph of Thoughts (Besta et al., 2024) further generalize the underlying topology, permitting non-linear dependencies such as aggregation and refinement. A key advantage of these methods is that verbal feedback can be applied at every thought level, which facilitates earlier identification of unproductive paths. However, exploring multiple states necessitates numerous large language model (LLM) calls, rendering these methods considerably more expensive than single-pass generation (Xie et al., 2023).

5 Pillar 3: Language as Learning Signal

The method above leave the model unchanged; we now examine how verbal feedback can persistently reshape the policy through training. As illustrated in Figure 5, the four subcategories form a compression spectrum: from feedback-conditioned modeling (§5.1), which preserves full critiques, through self-improvement (§5.2) and process supervision (§5.3), to preference shaping (§5.4), which reduces verbal judgement to a single scalar.

5.1 Feedback-Conditioned Modeling

At the upper end of the compression spectrum, feedback-conditioned modeling maintains verbal critiques in their entirety by conditioning the model directly on natural-language feedback during training. The training signal consists of a triple (x, v, a^*): the input, a critique of an initial response, and the revised output. A central question is whether to retain the critique within the training context. Early studies omitted the critique, using only the refined output as the supervision target (Scheurer et al., 2023). Subsequent work, such as ALT (Lloret et al., 2024), demonstrated that retaining feedback as conditioning context enables models to learn explicit mappings from critique to correction. Luo et al. (2025) further formalized this approach by treating free-form feedback as a primary conditioning signal with online bootstrapping. While preserving the full linguistic content of the critique allows these methods to retain the richest signal, this same receptiveness introduces a risk: models may learn to follow critiques indiscriminately, even when they are incorrect (Sharma et al., 2024).

5.2 Self-Improvement

Self-improvement approaches treat language as filtered text rather than as a conditioning context. The central mechanism is generate-then-filter: the model generates multiple candidate trajectories, verbal feedback signals determine which candidates are retained, and these selected outputs become supervised fine-tuning data for subsequent iterations (Singh et al., 2023). The filtering criterion directly shapes the model's learning. For example, in STaR (Zelikman et al., 2022), correctness of the final answer is used as the filter, while in Constitutional AI (Bai et al., 2022), adherence to safety principles serves this role. Explicit labeling is not always required; majority voting across diverse samples can identify likely correct traces (Huang et al., 2022a), and the model itself can act as a judge across iterations (Yuan et al., 2024). This self-contained loop can facilitate compound-

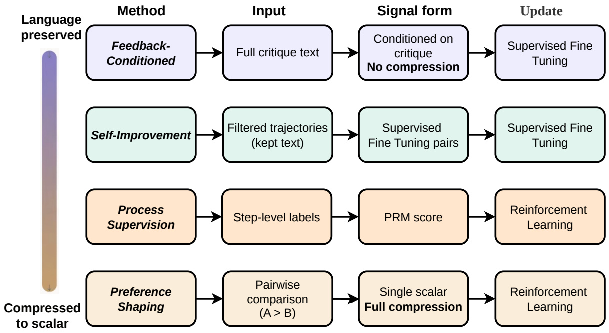

Figure 5: Training-time verbal feedback methods organized by degree of linguistic compression. Methods at the top preserve full natural-language feedback during training, while methods at the bottom compress it into scalar signals.

ing improvements over multiple cycles. However, filter quality remains a critical bottleneck: when the same model functions as both generator and judge, systematic biases may be amplified (Shumailov et al., 2023).

5.3 Process Supervision

Process supervision reduces verbal feedback to scalar scores at the step level. Annotators assess each intermediate reasoning step, and these evaluations are used to train a process reward model; the verbal content of the judgments is omitted. Despite this reduction, step-level supervision substantially outperforms outcome-level supervision (Lightman et al., 2024; Uesato et al., 2022), as localized feedback enables more precise credit assignment. The high cost of step-level annotation has motivated efforts to automate this process (Wang et al., 2024a; Luo et al., 2024). Generative process reward models that incorporate reasoning prior to scoring retain more verbal information while maintaining step-level granularity (Zhao et al., 2026; Liu et al., 2025b). Although step-level granularity offers significant advantages, it remains costly and is challenging to define in domains where intermediate correctness is ambiguous.

5.4 Preference Shaping

Preference shaping represents the maximal compression of verbal feedback, reducing a comparative judgment between two responses to a single scalar that guides policy updates. Building on the foundation of learning from verbal preferences (Christiano et al., 2017), InstructGPT (Ouyang et al., 2022) demonstrated large-scale implementation by training a reward model on pairwise rankings and optimizing the policy online. Direct Preference Optimization (DPO) (Rafailov et al., 2023) further simplified this process by eliminating the explicit reward model and training directly on preference pairs, using the comparison itself as the loss signal. Subsequent research has investigated additional alternatives (Ethayarajh et al., 2024; Hong et al., 2024a; Meng et al., 2024; Azar et al., 2024). Notably, Lee et al. (2023) replaced human annotators with AI judges, thereby enhancing the scalability of preference collection. The online and offline variants differ in practice: online optimization against a learned reward model can query feedback on fresh policy samples, while offline objectives such as DPO inherit the coverage of a fixed preference set. Although scalar compression reduces the explanatory richness of verbal feedback, the scalability enabled by llm generated judgments increasingly justifies this tradeoff. Under our inclusion criteria (Section 1), preference shaping marks the outer boundary of VRL: once the judgment is compressed to a preference bit before any learning, the linguistic structure is no longer consumed, so vanilla RLHF and DPO sit outside the paradigm. We retain them here as the endpoint of the compression spectrum because they clarify, by contrast, what the methods above preserve.

6 Insights, Discussion and Future Prospects

In practice, the three pillars operate together, as the coding-agent example in §2 illustrates. Five cross-cutting themes span them: algorithmic interactions and practitioner guidance (§6.1), feedback-model quality (§6.2), tool-interface design (§6.3), adversarial robustness (§6.4), and the absence of formal guarantees (§6.5).

6.1 Algorithmic Interactions and Practitioner Guidance

Different feedback forms pair with different optimization families, and the pairing carries assumptions a practitioner must check. Executable reward functions generated from language pair with online policy-gradient training; pairwise judgments pair with offline preference objectives such as DPO; filtered or feedback-conditioned text pairs with supervised fine-tuning; and step-level verbal judgments pair with process reward models (PRMs), which enable finer-grained credit assignment (Lightman et al., 2024). Each pairing has consequences inside the decision problem: rewards specified through language inherit objective mis-specification and reward-hacking risks; states represented through language raise information-loss and aliasing questions; and actions generated under LLM priors change exploration. Appendix B distills these considerations into a decision table keyed to properties a practitioner can assess before committing to a mechanism: verifiability of task outcomes, feedback latency and cost, and risk exposure such as reward hacking, sycophancy, and feedback poisoning.

6.2 Dedicated Feedback Models

Verbal feedback quality remains the key performance factor, but current practice typically reuses the same LLM for both generation and critique. Dedicated critics consistently outperform this approach: Shepherd (Wang et al., 2023b) provides more actionable feedback than untuned baselines, CriticGPT (McAleese et al., 2024) detects code bugs more reliably, and CritiqueLLM (Ke et al., 2024) scales critique data via multi-path prompting. Co-training critics with reasoners yields further improvements (Liu et al., 2025a), while generative process reward models that reason before scoring preserve more verbal signal than scalar PRMs (Khalifa et al., 2025). CriticBench (Lin et al., 2024) offers the first systematic benchmark for critique quality. A broader evaluation practice is emerging around reward-model benchmarks (Lambert et al., 2024) and step-level process benchmarks (Zheng et al., 2024); we recommend reporting feedback *quality* and feedback *utilization* separately, since models often fail to incorporate even accurate feedback (Jiang et al., 2025). These results indicate a natural next step: just as the field invested in instruction-tuned models (Ouyang et al., 2022), it should now develop feedback-tuned models whose objectives address error localization, actionability, and calibration. Recent advances in span-level feedback gradients (Wang et al., 2025b) and multi-turn didactic interactions (Klissarov et al., 2026) suggest such models can serve as differentiable training signals across each pillars.

6.3 Designing Tools for Verbal Feedback

Developer-oriented tools often yield outputs that are terse or conflate error types. Yang et al. (2024) demonstrated that LLM-optimized interfaces enhance agent performance, while Bandlamudi et al. (2025) showed ambiguous outputs cause many agent failures across 750 API calls. Tool-interactive critiquing requires interpretable output (Gou et al., 2024a), and executable code actions benefit agents by providing structured feedback (Wang et al., 2024b). RL on execution feedback (Gehring et al., 2024) presumes adequate tool-output signal, while agent harness frameworks (Ning et al., 2026) set tool quality as the limit of iterative refinement. However, a metric to measure this limit is lacking: tool-output compliance should quantify agent consumability. Tool APIs must expose structured metadata distinguishing task-level errors from infrastructure failures to enable agents to apply suitable recovery strategies.

6.4 Adversarial Verbal Feedback

Because VRL agents inherently act on feedback, adversarial feedback constitutes policy manipulation, introducing vulnerabilities beyond standard prompt injection. Zhan et al. (2025) demonstrated that adaptive attacks circumvent all eight tested defenses, while tool-level injection attains a 96.7% success rate on GPT-4o (Shi et al., 2025b). Hidden instructions in API responses covertly manipulate agents (Zhan et al., 2024), adversarial memory injection induces cross-session drift (Dong et al., 2026), and even benign incorrect feedback leads to sycophancy (Sharma et al., 2024). Defenses are developing: instruction hierarchies (Wallace et al., 2024), adversarial fine-tuning (Chen et al., 2025), and system-level design patterns (Beurer-Kellner et al., 2025) each mitigate narrow attack surfaces. Comprehensive solutions will require feedback-provenance mechanisms to verify source integrity and assign trust, as well as adversarial feedback benchmarks to evaluate robustness as the primary metric.

6.5 Theoretical Foundations

Verbal RL lacks formal guarantees; it remains unclear when verbal feedback enhances sample efficiency or policy quality. Three promising directions emerge. First, if verbal critics serve as noisy oracles with bounded error, PAC-learning frameworks (Strehl et al., 2009) can bound VRL sample complexity, and critic benchmarks (Lin et al., 2024) allow empirical estimation of error rates. Second, modeling feedback as partial observation links VRL to POMDP instruction following (Luketina et al., 2019), where belief-state planning yields convergence guarantees; McCallum et al. (2023) implement this with feedback-conditioned decision transformers. Third, rate-distortion theory may quantify signal retention from critique to scalar, clarifying when preference shaping aligns with feedback-conditioned methods. Empirical studies support these directions: language models learn from verbal feedback without scalar rewards (Luo et al., 2025), text feedback broadens RL capabilities (Song et al., 2026), and recent surveys link RLHF to language-conditioned RL theory (Kaufmann et al., 2023). Formalizing results in these areas would clarify when the cost of rich verbal feedback is warranted over scalar alternatives. We state these as open problems rather than established results. Concretely, the field's research agenda includes: the sample-efficiency conditions under which verbal feedback beats scalar reward; optimality and convergence when rewards are specified or shaped through language; temporal credit assignment from free-form critiques; provenance and robustness of feedback channels; and standardized, separate evaluation of feedback quality and feedback utilization. Each item is currently blocked by the absence of a formal account of what a verbal signal contributes beyond its scalarization.

7 Conclusion

In this paper, we present the first comprehensive survey of verbal reinforcement learning, organized around three pillars based on when language enters an agent's lifecycle: grounding signal, deliberative feedback, and learning signal. Together, these pillars reveal a common pattern: verbal feedback is rapidly becoming the primary medium for defining, updating, and improving agents.

Looking ahead, we anticipate that verbal feedback will become a first-class design primitive in agent architectures. The boundaries between pillars will increasingly blur as agents consume verbal feedback in unified loops that span grounding, deliberation, and learning within a single trajectory. The bottleneck will shift from *generating* feedback to *verifying* it, making feedback provenance, quality benchmarks, and adversarial robustness first-class infrastructure requirements. Finally, tool interfaces will need to be redesigned around agent consumability rather than human readability, as the gap between tool output quality and agent performance becomes the practical ceiling for iterative refinement.

Limitations

We aim to provide broad coverage of methods that use verbal feedback within the agent's lifecycle. Within each category, we focus on representative papers rather than attempting an exhaustive enumeration. Additionally, while our taxonomy offers a unifying framework, it may oversimplify methods that span multiple roles of language. Several approaches do not fit neatly into a single branch, since verbal feedback can simultaneously serve as a grounding signal, a deliberative mechanism, and a learning signal. Our taxonomy should therefore be understood as an organizing lens rather than a strict partition of the literature. Finally, our survey focuses primarily on methods in which verbal feedback is explicit and central to the agent's process, and therefore does not fully cover adjacent work where language plays a more auxiliary role. Language-based feedback itself carries failure modes that this survey can name but not yet bound: reward-specification failures when language compiles to the wrong objective, unreliable interpretation of feedback by the consuming model, weak grounding between verbal descriptions and environment state, and the transfer of language-guided policies to real-world settings. Societal impact follows the same channels: when feedback is incorrect, biased, adversarial, or poorly grounded, harm falls on users of high-stakes deployments such as coding assistants, robotics, clinical decision support, and education, and on the people affected by those systems' outputs. Sections 6.4 and 6.5 discuss mitigation directions. AI writing tools were used for language editing and proofreading only; all research contributions are the authors' own.

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A Comparison with Prior Surveys

This survey is, to our knowledge, the first organized around the verbal signal itself: which component of the decision problem a linguistic artifact modifies and when it takes effect. Prior overviews organize by mechanism (how a method works) or by agent capability (what a system does). Table 3 makes the positioning explicit. Relative to each row, our contribution is the signal-centric axis, the inclusion criteria of Section 1, the hybrid-case decision rules of Table 1, and the practitioner guidance of Appendix B.

B Choosing a Verbal-Feedback Mechanism

Table 4 distills the survey into prescriptive guidance. Each row states when a mechanism is the right choice, keyed to properties a practitioner can assess before building: whether task outcomes are verifiable, whether model weights are accessible, the latency and cost budget for feedback, and the dominant risk to monitor. Representative methods are drawn from the corresponding sections.

Table 3: Positioning against prior surveys and frameworks. Axis: the work's primary organizing principle. Feedback: signal types covered (S = scalar, P = preference, V = verbal). Stages: agent-lifecycle stages covered (D = problem definition, I = inference, T = training). Guide: whether prescriptive practitioner guidance is offered.

Survey / framework	**Axis**	**Feedback**	**Stages**	**Guide**
Luketina et al. (2019)	Language-conditional vs. language-assisted RL	S	D, T	✗
Pan et al. (2024)	When correction happens	V	I, T	✗
Kamoi et al. (2024)	When self-correction works	V	I	✗
Fernandes et al. (2023)	Feedback format and usage for generation	S, P, V	T	✗
Kaufmann et al. (2023)	RLHF pipeline	S, P	T	✗
Xiao et al. (2024)	Preference-optimization variants	P	T	✗
Zhang et al. (2025e)	Agent memory mechanisms	V	I	✗
Zhang et al. (2025b)	Test-time scaling (what/how/where)	S, V	I	✗
Gao et al. (2025)	What/when/how agents evolve	S, V	I, T	✗
Fang et al. (2025)	Agent-system optimizers	S, V	I, T	✗
Zhang et al. (2025a)	RL for agent capabilities	S	T	✗
Feng et al. (2024)	RL com-	V	I, T	✗

Table 4: Practitioner guidance: when to use language in each role.

Use language as	**When**	**Dominant risk**	**Representative**
Reward (compile to code or rubric)	Success is expressible as checks; simulator or executor available; training budget exists	Objective mis-specification; reward hacking	Eureka (Ma et al., 2024); Text2Reward (Xie et al., 2024)
State / task grounding	Raw observations too complex or task underspecified; language abstracts them faithfully	Information loss; state aliasing	ALFWorld (Shridhar et al., 2020); SayCan (Ahn et al., 2022)
Deliberative feedback	No weight access; latency budget allows iteration; a grounded critic (tool, tests) exists	Circular self-critique; inference cost	Self-Refine (Madaan et al., 2023); CRITIC (Gou et al., 2024a); ToT (Yao et al., 2023)
Memory	Tasks recur across episodes; lessons are reusable	Error persistence; memory poisoning	Reflexion (Shinn et al., 2023); ExpeL (Zhao et al., 2024); Voyager (Wang et al., 2023a)
Training signal	Persistent improvement needed; feedback plentiful; weights accessible	Sycophancy to wrong critiques; filter bias	FCP (Luo et al., 2025); STaR (Zelikman et al., 2022); PRMs (Lightman et al., 2024)